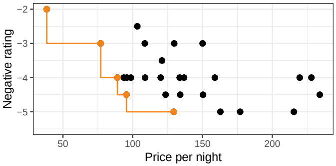
Optimization in Machine Learning

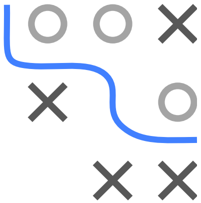
Multi-criteria Optimization
Introduction



Learning goals

- What is multi-criteria optimization?
- Motivating examples
- Pareto optimality

INTRODUCTORY EXAMPLE



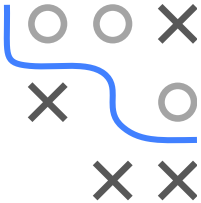
Often we want to solve optimization problems concerning several goals.
General applications:

- Medicine: maximum effect, but minimum side effect of a drug.
- Finances: maximum return, but minimum risk of an equity portfolio.
- Production planning: maximum revenue, but minimum costs.
- Booking a hotel: maximum rating, but minimum costs.

In machine learning:

- Sparse models: maximum predictive performance, but minimal number of features.
- Fast models: maximum predictive performance, but short prediction time.
- ...

INTRODUCTORY EXAMPLE

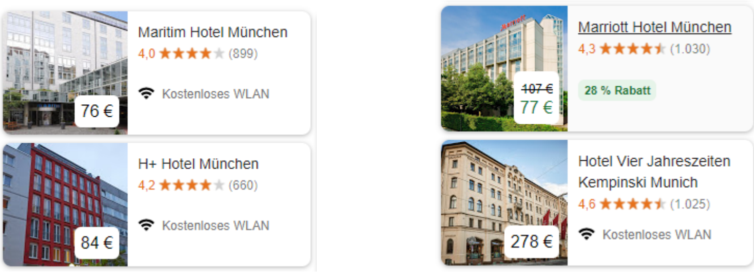


Example: Choose the best hotel by maximizing ratings subject to a maximum price per night. **Problems**:

- The result depends on how we select the maximum price; different price bounds give different solutions.
- We could also choose a minimum rating and optimize the price.
- The more objectives we include, the harder it gets to fix cutoffs like these.

Goal: Find a more general approach to solve multi-criteria problems.

INTRODUCTORY EXAMPLE



When booking a hotel: find the hotel with

- minimum price per night (**costs**), and
- maximum user rating (**performance**).

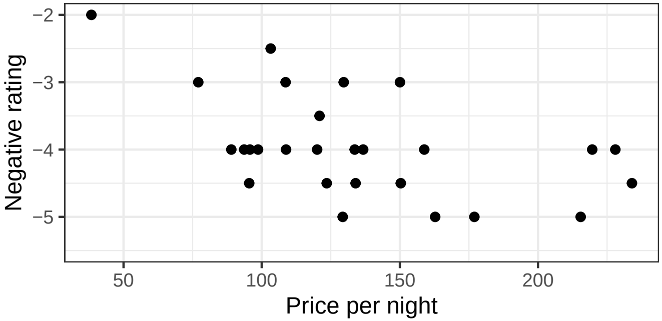
Since our standard is to minimize objectives, we minimize negative ratings.

INTRODUCTORY EXAMPLE

The objectives often conflict with each other:

- Lower price → usually lower hotel rating.
- Better rating → usually higher price.

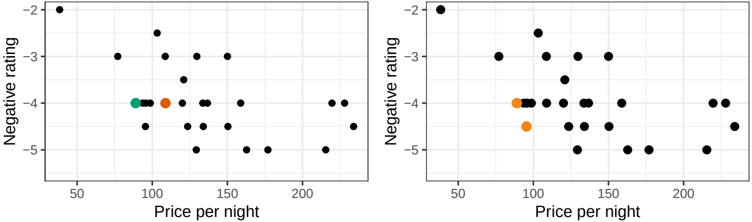
Example: (negative) average rating by hotel guests (1–5) vs. average price per night.



INTRODUCTORY EXAMPLE

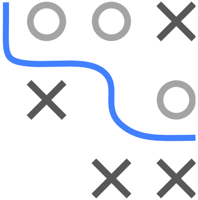
We cannot directly establish a total order on the objective space:

- Left: A hotel with rating 4 for 89 Euro ($f^{(1)} = (89, -4.0)$) is better than a hotel for 108 Euro with the same rating ($f^{(2)} = (108, -4.0)$).
- Right: How to decide if $f^{(1)} = (89, -4.0)$ is better or worse than $f^{(2)} = (95, -4.5)$?
- How much is one *rating point* worth?



Let $f = (\text{price}, -\text{rating})$. The candidate $f^{(1)} = (89, -4.0)$ dominates $f^{(2)} = (108, -4.0)$. That is, $f^{(1)}$ is at least as good in every dimension, and strictly better in one dimension. Hence $f^{(1)} \prec f^{(2)}$. But for $f^{(1)} = (89, -4.0)$ and $f^{(2)} = (95, -4.5)$ we cannot say either is strictly better.

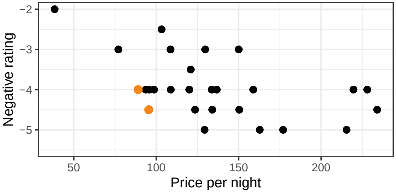
HOW TO DEFINE OPTIMALITY?

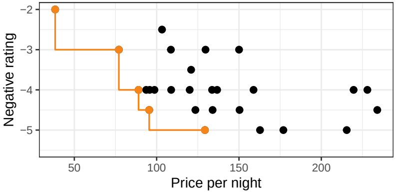


- We call these **incomparable** (or *equivalent* in the Pareto sense):

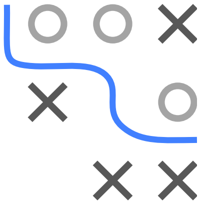
$$f^{(1)} \not\prec f^{(2)} \quad \text{and} \quad f^{(2)} \not\prec f^{(1)}.$$

- All such non-dominated points form the **Pareto front**.





DEFINITION: MULTI-CRITERIA OPTIMIZATION



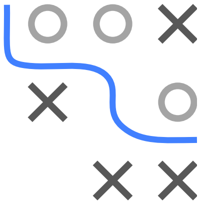
A **multi-criteria optimization problem** is defined by

$$\min_{\mathbf{x}\in\mathcal{S}} f(\mathbf{x}) \quad \Leftrightarrow \quad \min_{\mathbf{x}\in\mathcal{S}} \big(f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_m(\mathbf{x})\big),$$

with $\mathcal{S} \subset \mathbb{R}^n$ and a multi-criteria objective function $f: \mathcal{S} \to \mathbb{R}^m,\ m \geq 2$.

- **Goal:** minimize multiple target functions simultaneously.
- $(f_1(\mathbf{x}), \dots, f_m(\mathbf{x}))^\top$ maps each candidate $\mathbf{x}$ into the objective space $\mathbb{R}^m$.
- Typically no single best solution exists, as the objectives are often in conflict and cannot be totally ordered in $\mathbb{R}^m$.
- W.l.o.g. we always minimize.
- Also called multi-objective optimization, Pareto optimization, etc.

PARETO SETS AND PARETO OPTIMALITY



Definition: Given

$$\min_{\boldsymbol{x}\in\mathcal{S}} \big(f_1(\boldsymbol{x}), \dots, f_m(\boldsymbol{x})\big), \quad f_i : \mathcal{S} \to \mathbb{R}.$$

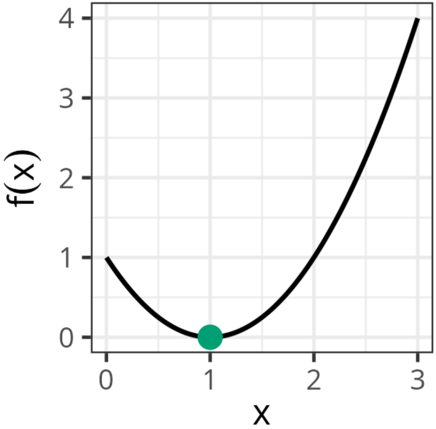
- A candidate $\boldsymbol{x}^{(1)}$ **(Pareto-)dominates** $\boldsymbol{x}^{(2)}$ if $f(\boldsymbol{x}^{(1)}) \prec f(\boldsymbol{x}^{(2)})$, i.e.:
  1. $f_i(\boldsymbol{x}^{(1)}) \leq f_i(\boldsymbol{x}^{(2)})$ for all i,
  2. $f_j(\boldsymbol{x}^{(1)}) < f_j(\boldsymbol{x}^{(2)})$ for at least one j.
- A candidate $\boldsymbol{x}^*$ not dominated by any other candidate is **Pareto optimal**.
- The set of all Pareto optimal candidates is the **Pareto set** $\mathcal{P} = \{\boldsymbol{x} \in \mathcal{S} \mid \nexists\, \tilde{\mathbf{x}} : f(\tilde{\mathbf{x}}) \prec f(\boldsymbol{x})\}$.
- $\mathcal{F} = f(\mathcal{P})$, the **Pareto front**, is the image of that set in objective space.

EXAMPLE: ONE OBJECTIVE FUNCTION

We consider

$$\min_x f(x) = (x-1)^2, \quad 0 \leq x \leq 3.$$

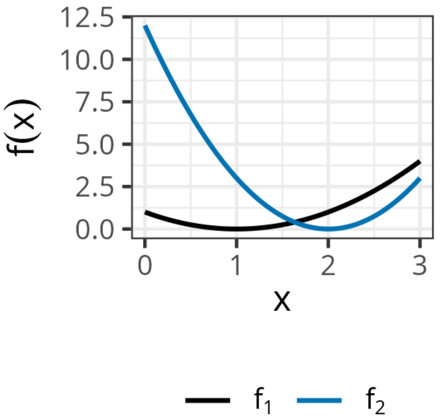
The optimum is at $x^* = 1$.

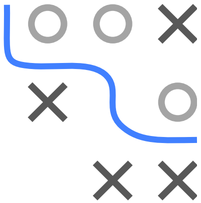


EXAMPLE: TWO TARGET FUNCTIONS

We extend the above problem to two objectives $f_1(x) = (x-1)^2$ and $f_2(x) = 3(x-2)^2$. Thus

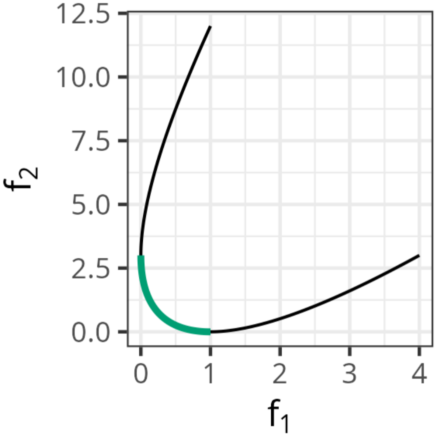
$$\min_x f(x) = \left(f_1(x),\, f_2(x)\right), \quad 0 \le x \le 3.$$





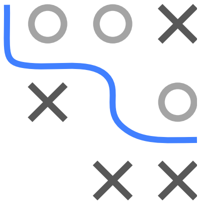
EXAMPLE: TWO TARGET FUNCTIONS

Visualizing both objectives: $\left(f_1(x),\ f_2(x)\right)$ for all $x \in [0, 3]$ in the objective space:



The Pareto front is shown in green.

A-PRIORI VS. A-POSTERIORI



- The Pareto set is a set of equally valid solutions.
- Often, we only want a **single** solution in practice.
- Without extra info, there is no unique best solution: one must choose based on further criteria (cost constraints, business preference, etc.). Two overall strategies:

- **A-priori approach**: incorporate user preferences *before* the optimization.
- **A-posteriori approach**: first find (an approximation of) the entire Pareto set, then let the user choose.

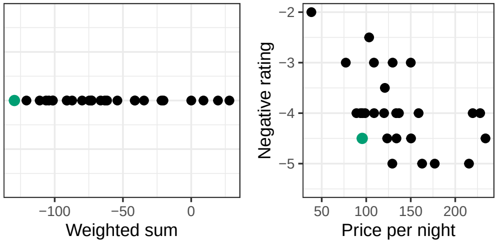
A-PRIORI PROCEDURE

Example: Weighted total

$$\min_{\boldsymbol{x}} \sum_{i=1}^{m} w_i f_i(\boldsymbol{x})$$

Prior knowledge: “One rating point is worth 50 Euro to this customer.”

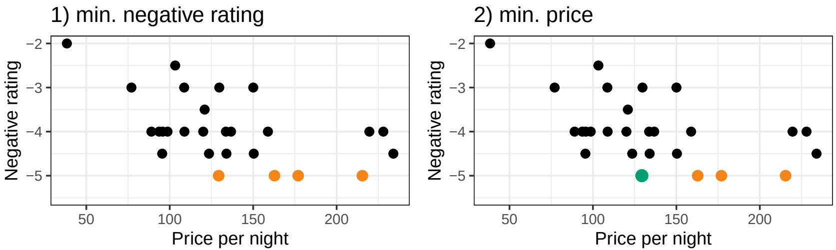
$$\min_{\text{Hotel}} \text{(Price)} \; - \; 50 \times \text{(Rating)}.$$



A-PRIORI PROCEDURE

Example: Lexicographic method

Prior knowledge: The rating is more important than the price. Hence we optimize the rating first, then the price among those best ratings:



A-PRIORI PROCEDURE

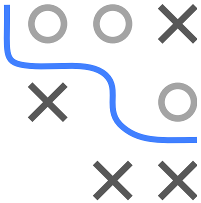
Lexicographic approach in general:

$$
\begin{aligned}
f_1^* &= \min_{\boldsymbol{x}\in\mathcal{S}} \; f_1(\boldsymbol{x}), \\
f_2^* &= \min_{\substack{\boldsymbol{x}\in\mathcal{S}:\\ f_1(\boldsymbol{x})=f_1^*}} \; f_2(\boldsymbol{x}), \\
f_3^* &= \min_{\substack{\boldsymbol{x}\in\mathcal{S}:\\ f_1(\boldsymbol{x})=f_1^*,\, f_2(\boldsymbol{x})=f_2^*}} \; f_3(\boldsymbol{x}), \\
&\;\;\vdots
\end{aligned}
$$

But different orderings yield different solutions.

Summary (a-priori):

- Implicit assumption: single-objective optimization is “easy.”
- We get only *one* solution, which depends strongly on the chosen weighting/order/etc.
- Varying these parameters systematically can produce several solutions, but may miss large parts of the non-dominated set.



A-POSTERIORI PROCEDURE

A-posteriori methods aim to

- find *all* optimal candidates (or a good approximation of them),
- let the user pick a single solution based on personal preferences or additional info (e.g. hotel location).

Hence, a-posteriori methods are more general.

Example: All Pareto-optimal hotels are shown on the left. User then picks from that subset on the right, based on hidden preferences (location, brand loyalty, etc.):